

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I Nagarajan Subhash / 192312489 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

N. Subhash
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	2	2	2	2	2
Analysis	20%	1	1	1	1	1
Code Quality	15%	1/2	1/2	1/2	1/2	1/2

28/12/24

✓

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

Short Answer

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Understanding	50%	Complete and correct understanding	Mostly correct understanding	Partial understanding	Incorrect understanding
Accuracy of Response	50%	Completely correct answer	Minor omission or error	Partially correct answer	Incorrect or irrelevant answer

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

IPv4

Given

IP

IPv4 Debugging and Optimization

Given

- IP Address 192.168.10.65
- Subnet mask : 255.255.255.192 (/26)
- Default Gateway : 192.168.10.1

Problem Identification

Subnet ranges:-

Network	Host Range	Broadcast
192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63
192.168.10.64/26	192.168.10.65 - 192.168.10.126	192.168.10.127

Debugging:-

Configured IP

192.168.10.65

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 - 192.168.10.126

Optimization :-

Assign a gateway within the same subnet

example

IP Address = 192.168.10.65

Subnet Mask = 255.255.255.192

Default Gateway = 192.168.10.126

Analysis

- * A host can communicate directly only with devices inside the same subnet.
- * Since the gateway was outside the subnet packets could not be forwarded.
- * After assigning a gateway within the same subnet, routing function correctly.

Usable Hosts

Host bits = 6

usable hosts

$$= 2^6 - 2$$

$$= 64 - 2$$

$$= 62 \text{ hosts}$$

IPv6 Debugging

Given

Device A

2001:db8:ff00:42::8329/64

Device B

2001:db8:0:0:1::1/64

Expanded IPv6 Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0000:0000:0000:0001

Network prefix

Device A

2001:db8:0:0::/64

Device B

2001:db8:0:0::/64

Debugging

The IPv6 addresses are in the same subnet.

- Incorrect gateway
- interface disabled

• Duplicate addresses

• Neighbour discovery failure

Optimization

Example

Router

Device A

2001:db8:0:0::1/64

Device B

2001:db8:0:0::20/64

Default gateway

2001:db8:0:0::1

Available hosts

IPv6 host bits

$$128 - 64 = 64$$

Hosts

$$2^{64}$$

= 18,446,744,073,709,551,616 addresses

IPv4 Subnet Optimization using VLSM

Subnet Details

Subnet	Network	Host Range	Broadcast	Usable Hosts
126	192.168.1.0	1-62	63	62
127	192.168.1.64	65-94	95	30
128	192.168.1.96	97-110	111	14

Unused Addresses

if department use

126 $\rightarrow$ 30 users

$$\text{Unused} = 62 - 30 = 32$$

127 $\rightarrow$ 15 users

$$\text{Unused} = 30 - 15 = 15$$

128 $\rightarrow$ 10 users

$$\text{Unused} = 14 - 10 = 4$$

Problem Identification

Large departments need more addresses

Small departments waste many addresses

Subnet allocation is inefficient

Debugging

Incorrect subnet sizes lead to

- IP wastage
- insufficient hosts
- Poor scalability.

5. Optimization using VLSM

Assign largest department first

Ex:-

HR $\rightarrow /26$

Sales $\rightarrow /27$

Finance $\rightarrow /28$

Analysis

- VLSM provides
- Better utilization
- Reduced wastage
- Easy expansion
- Improved network performance

IPv6 Debugging

Expanded Addresses

Device A

2001:0d68:0000:0000:0000:ff00:0042:8329

Device B

2001:0d68:0000:0000:0001:0000:0000:0001

Network prefix

2001:d68:0:0::/64

Same subnet

Yes

Debugging

check

- Gateway
- interface status
- Neighbor Discovery
- firewall
- Duplicate IPv6 Address

Optimized configuration

Gateway

2001::db8:0:0::1

Device A

2001::db8:0:0::10

Device B

2001::db8:0:0::20

Available hosts

2^{64}

= 18,446,744,073,709,551,616

Routing Table Debugging

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP

192.168.2.100

Longest prefix match

192.168.2.100 belongs to

192.168.2.0/24 ✓

Next Hop

10.0.0.3

Verification

Destination lies between

192.168.2.1

and

192.168.2.254

hence it belongs to correct subnet

Debugging

possible issues

- wrong next-hop address
- missing route
- interface down
- incorrect subnet mask

Optimization

Use the correct next-hop

192.168.2.0/24 → 10.0.0.3

Remove duplicate or incorrect routes and ensure
interfaces are active

Default route

if no matching destination exists,

Packets are forwarded to

10.0.0.1